

Retrieval-Augmented Generation for Academic Paper Question Answering

Abstract. We present PaperQA, a retrieval-augmented generation system for academic papers. It parses PDF documents into semantic chunks of 800 tokens with 10% overlap, retrieves via a hybrid of dense vectors and BM25 full-text search, reranks candidates with a cross-encoder, and generates citation-grounded answers. On 30 papers and 240 questions, answer accuracy improves from 62% to 85% after adding a hallucination constraint to the system prompt.

1 Introduction

Reading a single academic paper takes about two hours, and more than half of that time is spent locating key information. We target the single-paper question answering scenario and leave multi-paper comparison for future work. The core contribution is a three-layer diagnosis framework that attributes retrieval errors to the retrieval layer, the prompt layer, or the model layer. Experiments show 45% of errors come from the retrieval layer.

2 Method

We split each paper by section, then into chunks of at most 800 tokens with a 10% overlap between consecutive chunks. Chunk sizes of 400, 800 and 1200 tokens were compared; 800 gave the best retrieval recall. For retrieval we combine a bge-m3 dense vector index with BM25 full-text search, and apply query rewriting plus top-K reranking with bge-reranker-v2-m3. Every answer must cite the source page.

3 Results

After rewriting the system prompt to require answers grounded only in the provided context and to state explicitly when information is missing, answer accuracy rises from 62% to 85%. In a trial with 15 students, the average time to locate key information drops from 40 minutes to 10 minutes.

4 Conclusion

PaperQA demonstrates that citation-grounded generation and a strong hallucination constraint are the two most important factors for trustworthy academic question answering.